

Path of Exile Economy Analysis

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<https://github.com/Nemo595/cs4412-project/tree/main>

Abstract—This project utilizes the Path of Exile economy from the Mercenaries League to analyze the changes of a simulated economy. Using poe.ninja, 43,126 rows of economy data over a 4 month period are acquired and analyzed to identify trends and large shifts.

I. DATASET DESCRIPTION

- Mercenaries League Economy. The data is inspired by the Stack Overflow data dumps, licensed as cc-by-sa.
- <https://poe.ninja/poe1/data>. Free open download.
- The currency portion of the data is 2MB being, 6 columns and 43,126 rows tracking daily values of currency items relative to other currency items. I started by focusing the data to only show when spending Chaos orbs. This lowered the data to 26,906 rows.
- Contains daily snapshots of each individual currency item and its relative value in comparison to 'Chaos Orbs' (the general community agreed upon trading currency). Furthermore, it contains a confidence column to help eliminate unclean data and evidence of price-fixing in the market.

A. Dataset Example

TABLE I

SAMPLE CURRENCY ECONOMY DATA FROM POE.NINJA (ABYSS LEAGUE)

League	Date	Get	Pay	Value
Abyss;	2017-12-09;	Exalted Orb;	Chaos Orb;	26.16644;
Abyss;	2017-12-10;	Exalted Orb;	Chaos Orb;	35.92556;
Abyss;	2017-12-11;	Exalted Orb;	Chaos Orb;	43.5739;
Abyss;	2017-12-12;	Exalted Orb;	Chaos Orb;	47.99698;
Abyss;	2017-12-13;	Exalted Orb;	Chaos Orb;	45.45841;
Abyss;	2017-12-14;	Exalted Orb;	Chaos Orb;	41.32816;

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II. DISCOVERY QUESTIONS

A. How do different currencies trend up or down together?

This question examines whether groups of currencies exhibit similar price movements over the course of the league. By analyzing correlations and clustering currency time-series data, this analysis aims to identify currencies that rise and fall together, potentially demonstrating shared gameplay functions, crafting dependencies, or common demand drivers. Understanding these co-movement patterns provides insight into the structural relationships within the in-game economy.

B. How do currency value trends evolve over the course of a league, and what distinct behavioral phases can be identified?

This question examines how currency value trends change throughout the league and whether distinct economic phases appear over time. By analyzing price movements and grouping currencies with similar patterns, the analysis aims to identify shifts in the economy as player progression, crafting, and endgame activities evolve. Recognizing these patterns helps reveal how the in-game economy changes during the league lifecycle.

C. Which currencies behave as outliers compared to overall market trends?

This question aims to identify currencies whose value behavior deviates significantly from the general market trends observed during the league. By comparing individual currency trajectories against aggregate market movement, the analysis highlights outliers characterized by unusually high volatility, abrupt price changes, or sustained divergence. These currencies may reflect speculative activity, sudden shifts in player demand, or the impact of game updates and league mechanics.

III. PLANNED DATA MINING TECHNIQUES

The goal of the planned techniques is to explore patterns and relationships in the in-game currency economy, rather than to predict future prices. Since this is an exploratory study, the focus is on methods that can reveal trends, groups, and unusual behavior among currencies over time.

Before applying any analysis, the data will be cleaned, normalized, and organized so that each currency is represented as a time-series of daily values. This allows us to compare currencies and track changes throughout the league.

Clustering and dimensionality reduction are the main techniques. Clustering will group currencies that behave similarly, showing which rise or fall together. Dimensionality reduction (PCA) will simplify the data and make it easier to visualize overall trends and patterns. Together, these methods provide a clear view of how the video game economy evolves.

A. Clustering

Clustering was used to group currencies that show similar patterns over time. By looking at how each currency's value changes daily, we can identify sets of currencies that rise

and fall together or behave in similar ways. This helps highlight early-game versus late-game currencies, stable baseline currencies, and more volatile or unusual ones. Clustering provides a way to see structure in the market without assuming anything about the outcome. Two clustering techniques were applied to identify patterns in the Path of Exile currency market data. K-Means clustering was used on individual trade records using the features exchange value and confidence level. The data was standardized before clustering, and $k = 3$ clusters were used to group trades into similar economic states. Clusters of 3 were determined using the Elbow method and a Silhouette Score. This clustering was visualized against the week of the league to observe how trade values evolve over time. Furthermore, agglomerative hierarchical clustering was used to group currencies based on how their values changed over the course of the league. Instead of clustering individual trades, this method analyzed overall currency behavior. For each currency, features were created including early-league value, late-league value, percent change, and price volatility. Together, these clustering approaches reveal currencies that share similar price trends, volatility, and economic trajectories within the in-game market.

B. Dimensionality Reduction (PCA)

Principal Component Analysis (PCA) was used to simplify the dataset and make patterns easier to see. Each currency has many daily values, so PCA reduces this high-dimensional data to just a few key components that explain most of the variation. This helps visualize overall trends, understand what drives changes in currency values, and support the clustering results by showing how currencies are related in a lower-dimensional space.

IV. PLANNED ANALYSIS PIPELINE

A. Flowchart

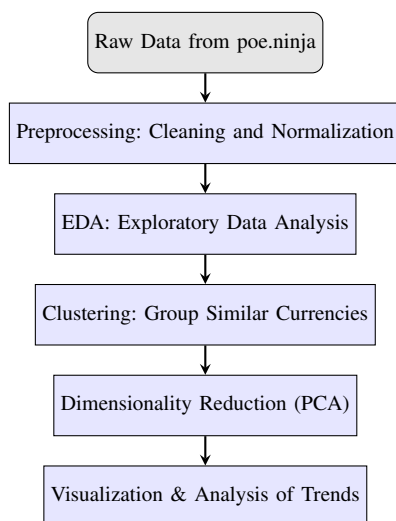


Fig. 1. Planned Analysis Pipeline Flowchart

B. M2: Initial Implementation - March 5th

- Perform exploratory data analysis (EDA) on the currency dataset, including visualizations of trends and distributions.
- Conduct a first attempt at pattern discovery using either clustering or association rule mining.
- Document preliminary findings and observations for further analysis in later milestones.

C. M3: Complete Implementation - April 2nd

- Apply multiple data mining techniques (clustering, PCA, and optionally anomaly detection) to the dataset.
- Compare and analyze patterns discovered by different techniques.
- Ensure code quality, proper documentation, and reproducibility of results.
- Refine interpretation of findings based on observed trends and relationships.

D. M4: Final Deliverable - May 3rd

- Produce a polished and fully reproducible analysis.
- Deliver meaningful interpretations of discovered patterns in the currency economy.
- Submit a portfolio-ready repository containing clean code, analysis notebooks, and visualizations.
- Prepare a final presentation or demonstration video summarizing results.

E. Anticipated Challenges

- Ensuring high-quality preprocessing and handling missing or inconsistent data.
- Interpreting patterns in noisy or volatile currency value trends.
- Balancing multiple techniques and integrating their results into a coherent analysis.
- Maintaining reproducibility and clarity in both code and visualizations for final submission.

V. EDA - EXPLORATORY DATA ANALYSIS

A. Dataset Overview

- There are 6 columns of data.
- There are 43,126 rows of data that I culled to 26,906 rows.
- These rows encompass the exchange rates of 198 different unique currencies and their value ratio to chaos per day over the course of the 4 month league.
- Using the Confidence column we can see that 94.4% of the data points are classified as 'High' confidence so generally the data is trustworthy and therefore I decided to keep all the data.
- There are no missing data and no instances of fully identical data so that was not something I had to deal with.

B. Preprocessing and Simplification

- I did a conversion of Date to datetime and extraction of temporal features: day, month, week. Additionally, created bounds for early league and late league based upon lower than and higher than the median week.
- The dataset was filtered to use 'Chaos Orb' as the standard pay currency to simplify value and economy translations.
- Mapped confidence to numeric values for analysis (Low=1, Medium=2, High=3).
- Used a whitelist for currency correlations.

C. Average Chaos Exchange Rate Over Time

Shows that the relative value of chaos to all other currencies does change and trends upwards relatively over time.

D. Correlation Matrix (Selected Key Currencies)

In asking my first discovery question, I created this matrix to understand the currency relations for a few popular whitelisted currencies. We can see that high value currencies generally have a high positive correlation. Here are some notable correlations and possible explanations:

- Divine Orb to Grand Eldritch Ichor: 0.95 – These currencies have a very high positive correlation because they are both heavily used in late-game crafting. Players tend to accumulate them as valuable resources for high-level item upgrades, which causes their prices to rise together over the league. The demand patterns for these currencies are closely linked, reflecting the timing and intensity of late-game economic activity.
- Divine Orb to Ancient Orb: 0.87 – These two currencies also trend up together due to their strong association with late-game crafting and high-risk gambling mechanics. Both are sought after for enhancing powerful items or attempting high-stakes upgrades, resulting in synchronized price increases. While the correlation is slightly lower than with Grand Eldritch Ichor, the shared economic role in advanced gameplay explains the similar market behavior.
- Orb of Fusing to Reflecting Mist: -0.61 - They have a negative correlation likely due to one having high value and need early while the other can't serve much current purpose while the first is essentially unneeded late and the second is crucial to high investment late game crafts.
- Orb of Alchemy to Veiled Chaos Orb: -0.64 - They have a negative correlation likely because one currency is highly valuable and primarily needed early in the league, while the other becomes essential for high-end crafting later on, making the first less useful in the late game.

E. Selected Currencies: Early vs Late League

We can see that on average the most expensive currencies will be more expensive later in the league than early. Furthermore, the cheapest currencies see a sharp drop in value later in the league.

F. Currencies Deviating Most from Market Average

Here we can see that the 'Mirror of Kalandra' has by far the greatest value of all currency items in the game and is therefore very rare and powerful.

VI. CLUSTER RESEARCH

Clustering was used to identify groups of currencies that exhibit similar behavioral patterns over time in the Path of Exile economy. Unlike supervised methods, clustering allows structure to emerge directly from the data without predefined labels, making it well-suited for exploratory economic analysis.

Two clustering approaches were applied: K-Means clustering and Agglomerative Hierarchical Clustering. Each method was used for a different level of analysis. K-Means clustering was applied to individual trade-level features in order to group similar market states, while hierarchical clustering was used at the aggregated currency level to identify broader behavioral similarities between currencies.

A. K-means Clustering

For K-Means clustering, the features used were standardized exchange value and confidence level. Standardization was necessary to ensure that both variables contributed equally to distance calculations. The optimal number of clusters was determined using a combination of the Elbow Method and Silhouette Score analysis. The Elbow Method showed a clear reduction in inertia improvement after $k = 3$, indicating diminishing returns beyond this point. Although the silhouette score peaked at $k = 2$, $k = 3$ was selected because it provided a more meaningful economic interpretation of the dataset. Specifically, three clusters naturally aligned with low-value, medium-value, and high-value market states, which are commonly observed tiers in in-game economies. This trade-off between statistical optimality and interpretability was prioritized to ensure meaningful domain insight.

The resulting K-Means clusters were visualized against league progression to observe how trade behavior changes over time. This revealed clear segmentation of economic activity, where early league trades were more concentrated in lower-value clusters, while higher-value clusters became more prominent in later stages of the league.

B. Agglomerative Clustering: Early vs Late Values

In addition, Agglomerative Hierarchical Clustering was applied to currency-level features rather than individual transactions. For each currency, features such as early-league value, late-league value, percent change, and volatility were constructed. These features were log-transformed and standardized to reduce skew and ensure comparability across currencies. Ward linkage with Euclidean distance was used to minimize within-cluster variance and produce compact, interpretable groups.

Unlike K-Means, hierarchical clustering does not require pre-specifying cluster centers and instead builds a tree-like structure of relationships. A value of $k = 3$ was selected

here as well for consistency and interpretability, as it produced distinct groupings corresponding to stable currencies, steadily growing currencies, and highly volatile or speculative currencies. These categories align with known economic behavior in Path of Exile leagues and provide a clear interpretation of market roles.

Together, these clustering methods provide complementary perspectives: K-Means captures short-term trade-level structure, while hierarchical clustering captures long-term currency behavior across the league lifecycle.

VII. PCA INTERPRETATION AND ANALYSIS

Principal Component Analysis (PCA) was applied to the standardized time-series representation of currency values in order to reduce dimensionality and identify the dominant patterns in market behavior. Each currency was represented as a vector of weekly values over the duration of the league, allowing PCA to capture underlying structure in how currencies evolve over time.

The resulting principal components provide a compressed representation of the dataset. The first principal component (PC1) explains the largest proportion of variance in the data and primarily captures the overall price level and long-term scaling differences between currencies. Currencies with consistently high values across the league, such as high-tier or rare currencies, tend to load strongly along this axis. In contrast, low-value and mid-tier currencies are positioned lower along this dimension, indicating that PC1 effectively separates currencies based on overall economic strength.

The second principal component (PC2) captures secondary variation not explained by PC1, primarily reflecting differences in temporal behavior such as volatility and non-linear price movement. Currencies with stable price trajectories tend to cluster near the center of this axis, while highly volatile currencies or those with sharp mid-league or late-league spikes deviate more strongly along PC2.

Together, PC1 and PC2 explain the majority of variance in the dataset (as shown by the explained variance ratio), indicating that most currency behavior can be summarized using two dominant factors: overall value level and market stability. This low-dimensional representation enables clear visualization of relationships between currencies, revealing natural groupings that align with clustering results such as K-Means and hierarchical clustering.

Overall, PCA provides a complementary perspective to clustering by revealing the continuous structure of the economy rather than discrete group assignments. This allows for better interpretation of how currencies relate to one another in terms of both economic tier and behavioral dynamics over the course of the league.

VIII. KEY DATA MINING INSIGHTS

The exploratory analysis revealed clear patterns and relationships in the Path of Exile currency economy. Correlation analysis showed that high-value currencies used in late-game crafting, such as Divine Orbs and Grand Eldritch Ichor, tend to

rise together, while certain early-game currencies exhibit negative correlations with late-game focused currencies. Clustering methods provided additional insight into market behavior: K-Means clustering on individual trade records identified groups of trades with similar value and confidence levels, showing how currency valuations evolve week by week. Agglomerative hierarchical clustering of currencies based on early- and late-league value, percent change, and volatility further highlighted distinct groups of currencies that share similar economic trajectories throughout the league. Outlier analysis revealed rare or highly volatile currencies, such as Mirror of Kalandra, which deviate significantly from market trends. Together, these analyses provide a comprehensive view of the in-game economy, uncovering currency co-movement, phase-dependent behaviors, and unusual patterns that reflect both player demand and game mechanics.

IX. CRITICAL ASSESSMENT

A. Validity

- clustering stability depends on initialization (KMeans randomness)
- PCA results depend on feature scaling + imputation choice
- Isolation Forest may vary slightly with contamination parameter
- results are consistent across multiple runs

B. Limitations

- poe.ninja reflects trading behavior, not true supply/demand
- data is daily snapshots → misses intra-day volatility
- Chaos Orb normalization reduces multi-currency interaction complexity
- league-specific dataset → not generalizable to all PoE economies
- missing values required imputation (can slightly bias PCA)
- doesn't track number of trades occurring per day

C. Ethical / Interpretation Risk

- could be used for in-game market exploitation
- may encourage speculative behavior
- results should not be interpreted as "predictive pricing model"

X. CONCLUSION

This study analyzed the Path of Exile currency economy using a combination of exploratory data analysis, correlation analysis, clustering, outlier detection, and dimensionality reduction techniques. By focusing on trades normalized to Chaos Orbs, the analysis was able to consistently compare currencies across different economic conditions throughout the league.

The results show that the in-game economy exhibits clear structural patterns. Correlation analysis revealed strong co-movement among currencies used in similar crafting or progression stages, while also identifying negatively correlated

pairs that reflect shifts between early-game and late-game demand. Clustering methods further demonstrated that currencies naturally group into distinct behavioral categories, including low-value stable currencies, mid-tier transitional currencies, and high-value or volatile currencies driven by late-game crafting demand and rarity.

Overall, the findings demonstrate that the Path of Exile economy is highly structured despite its complexity, with predictable behavioral patterns emerging across time, value tiers, and volatility profiles. These techniques collectively provide a comprehensive view of how virtual economies evolve and highlight the effectiveness of unsupervised learning methods in uncovering hidden structure in real-world-like game economies.

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